

Phase 1 Task Breakdown: Cross-Device AI Assistant

Every task needed to produce the Phase 1 report. No code in this phase. Choose tasks that add up to your share of points, then send your list to the team lead.

~18

points per member

aim for 16–20 · 91 points in total

The system we are designing

One AI assistant runs locally on the user's strongest device. It connects their PC, laptop, phone and iPad peer-to-peer over the local network, with no internet and no cloud server, and understands files and data across all of them.

Features: duplicate detection · unified search · password management · clipboard sync · screen sharing with remote control · extra screen · direct file sharing · offline device records.

What Phase 1 must deliver (from the supervisor's sample reports)

- 1. Introduction:** overview, scope, objectives, motivation, plan
- 2. Literature Review:** background, related systems, contribution
- 3. Problem Analysis:** problem, functional & non-functional requirements, evaluation plan
- 4. System Design:** architecture, DFD (Level 0 & 1), sequence diagrams, use case diagram

Plus abstract, references, lists of tables and figures.

Points = effort: 1 a few hours 2 1–2 days 3 3–5 days 4 about a week 5 1–2 weeks

A. Report writing Chapters 1–3 and front/back matter · 23 points

ID	TASK	WHAT IT INCLUDES	POINTS
A1	Chapter 1: Introduction	Introduction, project overview, scope, aim and objectives, motivation, report outline.	2
A2	Project plan & schedule	Schedule tables for weeks 1–17, kept up to date.	1
A3	Abstract	One-page abstract, written last.	1
A4	Chapter 2: Background	The concepts we rely on: P2P networking, on-device AI, duplicate detection methods, password encryption, screen streaming.	3
A5	Chapter 2: Existing related systems	Review similar apps and papers (e.g. Syncthing, KDE Connect, LocalSend, Apple Continuity, Bitwarden, Spacedesk, dupeGuru) with a comparison table.	4
A6	Chapter 2: Contribution	The gap our system fills.	1
A7	Chapter 3: Problem specification	Clear statement of the problem and who it affects.	1
A8	Chapter 3: Functional requirements	Collect every feature's requirements from the section C owners, number them and make them consistent.	2
A9	Chapter 3: Non-functional requirements	Privacy, security, performance, offline use, battery, cross-platform support, usability.	2
A10	Implementation & evaluation plan	How each feature will be built and tested in Phase 2, with success criteria.	3
A11	References & final assembly	IEEE references, contents and figure lists, formatting, proofreading the whole report.	3

B. Overall system design Chapter 4, the whole system · 20 points

ID	TASK	WHAT IT INCLUDES	POINTS
B1	System architecture	The main architecture diagram and its explanation: device app, P2P layer, AI node, feature modules, local storage, interface. Everyone's design builds on it.	5
B2	DFD Level 0 (context)	Context diagram and external entities table.	2
B3	DFD Level 1	All main processes and data flows, and the processes table.	4
B4	Use case diagram	All actors and use cases, with a short description of each.	3
B5	Data model	What the system stores and how it relates: devices, file index, duplicates, vault, sessions.	3
B6	Technology & platform choices	Languages, frameworks, AI model and runtime, P2P library, supported platforms.	3

C. Feature design Each: how it works, a sequence diagram, design decisions, its requirements · 38 points

ID	TASK	WHAT IT INCLUDES	POINTS
C1	Device discovery, pairing & trust	How devices find each other, how a new device joins (e.g. QR code), and how they trust each other. Every feature depends on this.	5
C2	AI assistant node	Choosing the strongest device, which local model to run, what the assistant does, and what happens when that device goes offline.	5
C3	File indexing & offline records	How each device lists its files, how the lists are shared, and how an offline device's last state is kept.	3
C4	Duplicate file detection	Exact copies (hashing) and similar photos and videos (AI), and how results are shown. <i>Choosing which device keeps the file: to be confirmed.</i>	5
C5	Unified search	Search every device from any device, offline ones included.	2
C6	Direct file sharing	Sending files between devices: transfer flow, large files, interruptions.	2
C7	Password management	Collecting accounts per device, the encrypted vault, syncing it, and threats to guard against.	4
C8	Clipboard sync	Copy on one device, paste on another (text and images).	1
C9	Screen sharing & remote control	Streaming a screen, sending mouse and keyboard back, and each platform's permissions.	4
C10	Extra screen	Using the iPad as a second monitor, with a study of what iOS allows.	4
C11	Security & privacy design	Encryption between devices, access control, where data is stored, lost-device protection.	3

D. User interface design 6 points

ID	TASK	WHAT IT INCLUDES	POINTS
D1	Screen wireframes	Main screens for desktop and mobile: devices, duplicates, search, vault, sharing, screen tools.	4
D2	User flows	Step-by-step journeys: setup and pairing, cleaning duplicates, finding and sending a file.	2

E. Coordination 4 points

ID	TASK	WHAT IT INCLUDES	POINTS
E1	Team & supervisor coordination	Meetings with the supervisor, meeting notes, tracking the schedule.	2
E2	Phase 1 presentation	Slides for the Phase 1 discussion.	2

How to choose: pick tasks totalling about 18 points and send the task IDs to the team lead.

Upload your work to: github.com/mshraky3/graduation-project-phase-one